MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, VOLTAGE-TO-FREQUENCY CONVERTERS,

MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic voltage-to-frequency converters. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device types. The device types shall be as follows:

Device type	Circuit
01	Voltage-to-frequency converter
02	Voltage-to-frequency converter
03	Voltage-to-frequency converter

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package
I	A-2 (10-lead can)
2	C-2 (20-terminal .350" x .350"), square chip carrier package

1.3 Absolute maximum ratings.

Positive supply voltage ($+V_{CC}$) to common:	
Device types 01 and 03	+18 V
Device type 02	+20 V
Negative supply voltage ($-V_{CC}$) to common:	
Device types 01 and 03	-18 V
Device type 02	-20 V
Input voltage range (V_{IN})	$\pm V_{CC}$
Differential input voltage range	±10 V

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center, (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Output pull-up supply voltage (V_{PU}):	
Device types 01 and 02	0 V to 30 V 1/
Device type 03	0 V to 36 V 1/
Output sink current (pin 6):	
Device type 01	16 mA
Device types 02 and 03	50 mA
Comparator input voltage	$-V_{CC}$ to $+V_{CC}$
Output current:	
Device type 01	-20 mA, +0.75 mA
Device type 02	-20 mA
Device type 03	-50 mA, +0.75 mA
Storage temperature range	-65°C to +150°C
Lead soldering temperature:	
Cases C and I (60 s)	300°C
Case 2 (10 s)	300°C
Junction temperature (T_J)	+175°C

1.4 Recommended operating conditions.

Positive supply voltage range ($+V_{CC}$)	(01) +11.0 V to +17.0 V (02) +13.0 V to +19.0 V (03) +10.8 V to +17.0 V
Negative supply voltage range ($-V_{CC}$)	(01) -11.0 V to -17.0 V (02) -13.0 V to -19.0 V (03) -10.8 V to -17.0 V
Output pull-up supply (V_{PU})	+4.5 V to +17 V 1/
Negative input voltage range ($-V_{IN}$)	-10 V to 0 V
Input current range, (max)	
pin 2	(01, 02) 0.5 mA 2/
pin 2 or 14	(02) 250 µA 3/
pin 3	(03) 0.5 mA
Full scale frequency range	(01) 0 to 200 kHz (02, 03) 0 to 1 MHz
Recommended duty cycle at full scale frequency	f<100 kHz (01, 02, 03) 25% f>100 kHz (01, 02, 03) 50%
Ambient operating temperature range	-55°C to +125°C

1.5 Power and thermal characteristics.

Package	Case outline	Maximum allowable power dissipation	Maximum θJC	Maximum θJA
10-lead can	A-2	225 mW at T_A = +125°C	70°C/W	150°C/W
14-ceramic DIP	D-1	400 mW at T_A = +125°C	50°C/W	120°C/W
20-terminal CCP	C-2	225 mW at T_A = +125°C	55°C/W	120°C/W

1/ V_{PU} is the supply voltage connected to pin 6 via R2. See figure 5.

2/ 50% duty cycle.

3/ 25% duty cycle.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard, form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Case outlines. The case outlines shall be as specified in 1.2.3 herein.

3.2.2 Circuit diagrams and terminal connections. The circuit diagrams and terminal connections shall be as specified on figure 1.

3.2.3 Functional block diagram. The functional block diagram shall be as specified on figure 2.

3.2.4 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request.

3.2.5 Package and sealing material. Package and sealing material shall be in accordance with MIL-M-38510 (see 6.5).

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

Text continues on page 32

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TABLE I. Electrical performance characteristics for device type 01.

Test	Symbol	Conditions (see 3.4 and figure 4) unless otherwise specified	Limits Min	Limits Max	Units 1/ 4/
Quiescent current	I_{CC}		0	8.0	mA
	I_{EE}		-8.0	0	
Input bias current	$+I_{IB}$	T_A = +25°C	-150	150	nA
	$-I_{IB}$	T_A = +25°C	-90	90	
Differential input resistance	R_{IND}	T_A = +25°C 2/	330	---	kΩ
Differential input capacitance	C_{IND}	T_A = +25°C 2/	10	---	pF
Common mode resistance	R_{INCM}	T_A = +25°C 2/	300	---	MΩ
Common mode	C_{INCM}	T_A = +25°C 2/	3	---	pF
Low level output voltage 3/	V_{OL1}	I_{SINK} = 8 mA T_A = +25°C	0	0.4	V
		I_{SINK} = 5 mA -55°C ≤ T_A ≤ +125°C	0	0.4	
	V_{OL2}	I_{SINK} = 16 mA T_A = +25°C	0	0.8	
Leakage current	I_{OH}	LOGIC "1" V_O = 17 V (pin 6) T_A = +25°C	---	1	µA
Linearity error 4/	L_E	Positive input voltage 10 Hz ≤ f ≤ 10 kHz	-0.01	0.01	% of FSR
		Positive input voltage 200 Hz ≤ f ≤ 200 kHz	-.1	.1	
Offset error	V_{OS}	T_A = +25°C f = 10 kHz	-4	4	mV
		f = 200 kHz	-4	4	
Offset error drift	dV_{OS}/dT	-55°C ≤ T_A ≤ +125°C f = 10 kHz	-30	30	uV/°C
		f = 200 kHz	-30	30	
Gain error 5/	A_E	T_A = +25°C f = 10 kHz	-10	10	% of FSR
		f = 200 kHz	-10	10	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 01 - Continued.

Test	Symbol	Conditions (see 3.4 and figure 4) unless otherwise specified		Limits Min	Limits Max	Units 1/ 4/
Gain error drift 5/	d_{AE}/dT	$-55^{\circ}C \leq T_A \leq +125^{\circ}C$	f = 10 kHz	-200	200	ppm/°C
			f = 200 kHz	-200	200	
Full scale drift (offset drift + gain drift) 6/	$d_{F(FS)}/dT$	f = 10 kHz	$-55^{\circ}C \leq T_A \leq +25^{\circ}C$	-200	120	ppm of FSR/°C
			$+25^{\circ}C \leq T_A \leq +125^{\circ}C$	-120	200	
Full scale drift (offset drift + gain drift) 6/	$d_{F(FS)}/dT$	f = 200 kHz	$-55^{\circ}C \leq T_A \leq +25^{\circ}C$	-400	100	ppm of FSR/°C
			$+25^{\circ}C \leq T_A \leq +125^{\circ}C$	-150	150	
Power supply sensitivity	P_{SS1}	$11.0\ V \leq V_{CC} \leq 17.0\ V$ $T_A = +25^{\circ}C$		-.04	.04	% of FSR /% supply
	P_{SS2}	$-17.0\ V \leq V_{EE} \leq -11.0\ V$ $T_A = +25^{\circ}C$		-.04	.04	
Settling time	t_S	V/F to specified linearity V_{IN} = 10 V 2/ 6/ 7/		---	---	---
Overload recovery		50% overload 2/ 6/ 7/		---	---	---

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 02.

Test	Symbol	Conditions (see 3.4 and figure 4) unless otherwise specified	Limits Min	Limits Max	Units 1/ 4/
Quiescent current	I_{CC}		0	8.0	mA
	I_{EE}		-8.0	0	
Input bias current	$+I_{IB}$	T_A = +25°C	-30	30	nA
	$-I_{IB}$	T_A = +25°C	-8	-8	
Differential input resistance	R_{IND}	T_A = +25°C 2/	300	---	kΩ
Differential input capacitance	C_{IND}	T_A = +25°C 2/	5	---	pF
Common mode resistance	R_{INCM}	T_A = +25°C 2/	300	---	MΩ
Common mode	C_{INCM}	T_A = +25°C 2/	3	---	pF
Low level output voltage 3/	V_{OL1}	I_{SINK} = 8 mA T_A = +25°C	0	0.4	V
		I_{SINK} = 5 mA -55°C ≤ T_A ≤ +125°C	0	0.4	
	V_{OL2}	I_{SINK} = 16 mA T_A = +25°C	0	0.8	
Leakage current	I_{OH}	LOGIC "1" V_O = 19 V (pin 6) T_A = +25°C	---	1	µA
Linearity error 4/	L_E	Positive input voltage 10 Hz ≤ f ≤ 10 kHz	-.005	.005	% of FSR
		Positive input voltage 1 Hz ≤ f ≤ 1 MHz	-.1	.1	
Offset error	V_{OS}	T_A = +25°C f = 10 kHz	-0.5	0.5	mV
		f = 1 MHz	-0.5	0.5	
Offset error drift	dV_{OS}/dT	-55°C ≤ T_A ≤ +125°C f = 10 kHz	-5	5	µV/°C
		f = 1 MHz	-5	5	
Gain error 5/	A_E	T_A = +25°C f = 10 kHz	-10	10	% of FSR
		f = 1 MHz	-10	10	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 02 - Continued.

Test	Symbol	Conditions (see 3.4 and figure 4) unless otherwise specified		Limits Min	Limits Max	Units 1/ 4/
Gain error drift 5/	d_{AE}/dT	$-55^{\circ}C \leq T_A \leq +125^{\circ}C$	f = 10 kHz	-50	50	ppm/°C
			f = 1 MHz	-50	50	
Full scale drift (offset drift + gain drift) 6/	$d_{F(FS)}/dT$	f = 10 kHz	$-55^{\circ}C \leq T_A \leq +25^{\circ}C$	-50	50	ppm of FSR/°C
			$+25^{\circ}C \leq T_A \leq +125^{\circ}C$	-50	50	
Full scale drift (offset drift + gain drift) 6/	$d_{F(FS)}/dT$	f = 1 MHz	$-55^{\circ}C \leq T_A \leq +25^{\circ}C$	-50	50	ppm of FSR/°C
			$+25^{\circ}C \leq T_A \leq +125^{\circ}C$	-100	0	
Power supply sensitivity	P_{SS1}	$13.0\ V \leq V_{CC} \leq 19.0\ V$ $T_A = +25^{\circ}C$		-.025	.025	% of FSR /% supply
	P_{SS2}	$-19.0\ V \leq V_{EE} \leq -13.0\ V$ $T_A = +25^{\circ}C$		-.025	.025	
Settling time	t_S	V/F to specified linearity V_{IN} = 10 V 2/ 7/		---	---	---
Overload recovery		50% overload 2/ 7/		---	---	---

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 03.

Test	Symbol	Conditions (see 3.4 and figure 4) unless otherwise specified	Limits Min	Limits Max	Units 1/ 4/
Quiescent current	I_{CC}		0	8.0	mA
	I_{EE}		-8.0	0	
Input bias current	$+I_{IB}$	T_A = +25°C	-100	100	nA
	$-I_{IB}$	T_A = +25°C	-20	20	
Differential input resistance	R_{IND}	T_A = +25°C 2/	2	---	kΩ
Differential input capacitance	C_{IND}	T_A = +25°C 2/	10	---	pF
Common mode resistance	R_{INCM}	T_A = +25°C 2/	100	---	MΩ
Common mode	C_{INCM}	T_A = +25°C 2/	10	---	pF
Low level output voltage 3/	V_{OL1}	I_{SINK} = 8 mA T_A = +25°C	0	0.4	V
		I_{SINK} = 5 mA -55°C ≤ T_A ≤ +125°C	0	0.4	
	V_{OL2}	I_{SINK} = 16 mA T_A = +25°C	0	0.8	
Leakage current	I_{OH}	LOGIC "1" V_O = 17 V (pin 8) T_A = +25°C	---	100	µA
Linearity error 4/	L_E	Positive input voltage 10 Hz ≤ f ≤ 10 kHz	-.005	.005	% of FSR
		Positive input voltage 1 kHz ≤ f ≤ 1 MHz	-.1	.1	
Offset error	V_{OS}	T_A = +25°C f = 10 kHz	-4	4	mV
		20 mV ≤ V_{IN} ≤ 10 V f = 1 MHz	-4	4	
Offset error drift	dV_{OS}/dT	-55°C ≤ T_A ≤ +125°C f = 10 kHz	-30	30	µV/°C
		f = 1 MHz	-30	30	
Gain error 5/	A_E	T_A = +25°C f = 10 kHz	-10	10	% of FSR
		f = 1 MHz	-15	15	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics for device type 03 - Continued.

Test	Symbol	Conditions (see 3.4 and figure 4) unless otherwise specified	Limits Min	Limits Max	Units 1/ 4/
Gain error drift 5/	d_{AE}/dT	$-55^{\circ}C \leq T_A \leq +125^{\circ}C$ f = 10 kHz	-150	150	ppm/°C
		f = 1 MHz	-350	350	
Full scale drift (offset drift + gain drift) 6/	$d_{F(FS)}/dT$	f = 10 kHz $-55^{\circ}C \leq T_A \leq +25^{\circ}C$	-75	75	ppm of FSR/°C
		$+25^{\circ}C \leq T_A \leq +125^{\circ}C$	-75	75	
Full scale drift (offset drift + gain drift) 6/	$d_{F(FS)}/dT$	f = 1 MHz $-55^{\circ}C \leq T_A \leq +25^{\circ}C$	-150	150	ppm of FSR/°C
		$+25^{\circ}C \leq T_A \leq +125^{\circ}C$	-150	150	
Power supply sensitivity	P_{SS1}	$10.8\ V \leq V_{CC} \leq 17.0\ V$ $T_A = +25^{\circ}C$	-.02	.02	% of FSR /% supply
	P_{SS2}	$-17.0\ V \leq V_{EE} \leq -10.8\ V$ $T_A = +25^{\circ}C$	-.02	.02	
Bipolar offset error 8/	B_{pvos}	$T_A = +25^{\circ}C$ f = 10 kHz	-500	500	Hz
		$-4.98\ V \leq V_{IN} \leq 5.0\ V$ f = 1 MHz	-50	50	kHz
Bipolar offset drift 8/	dB_{pvos}/dT	$-55^{\circ}C \leq T_A \leq +125^{\circ}C$	-.02	.02	% of FSR/°C
Settling time	t_S	V/F to specified linearity V_{IN} = 10 V 2/ 7/	---	---	---
Overload recovery		50% overload 2/ 7/	---	---	---

1/ FSR = full scale range (corresponds to full scale frequency or full scale input voltage).

2/ These parameters are for information only.

3/ Open collector input.

4/ Linearity error is specified at any operating frequency as the deviation from a straight line intersecting 100 percent of full scale frequency and 0.1 percent of full scale frequency.

5/ Adjustable to zero.

6/ Positive drift is defined to be increasing frequency with increasing temperature.

7/ 1 µs plus one pulse (after 1 µs) of new frequency.

8/ 1.24 kΩ resistor between pins 4 and 5, one percent metal film.

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3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended ambient operating temperature range, unless otherwise specified. The following conditions shall also apply unless otherwise specified:

$+V_{CC}$ = +15.0 V

$-V_{CC}$ = -15.0 V

$-55^{\circ}C \leq T_A \leq +125^{\circ}C$

V_{PU} = +5.0 V

3.4.1 Offset and gain error null. (01) The voltage-to-frequency converter is capable of being nulled to zero offset and zero gain error using the circuits shown on figures 5 and 6. R_5 affects zero offset error; R_3 affects gain error. C_1 and C_2 are ±50 ppm teflon capacitors.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

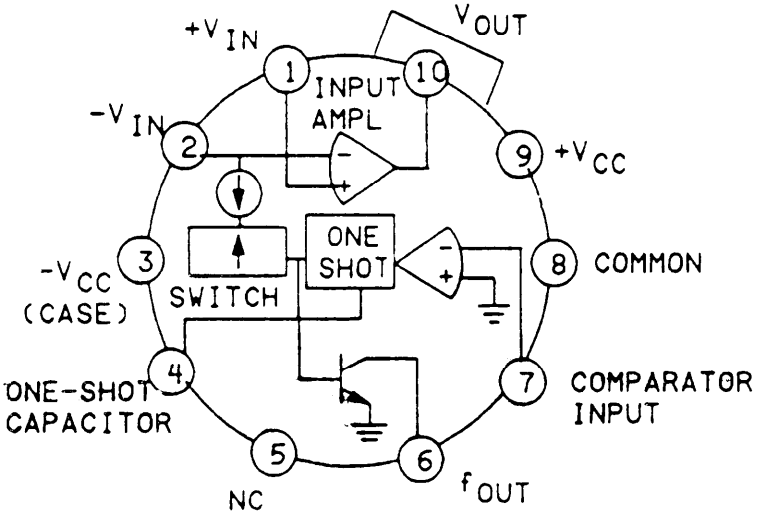
TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters 1/ (method 5004)	1,2,3	1,2,3
Group A test requirements (method 5005)	1,2,3	1,2,3
Group C end-point and group B, class S, electrical parameters (method 5005)	1 and table IV delta limits	1 and table IV delta limits
Additional electrical subgroups for group C periodic inspections	N/A	---
Group D end-point electrical parameters (method 5005)	1,2,3	1

1/ PDA applies to subgroup 1.

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Device type 01



(TOP VIEW)

NC = NO INTERNAL CONNECTION.
EXTERNAL CONNECTION PERMITTED.

Device type 02

Case I

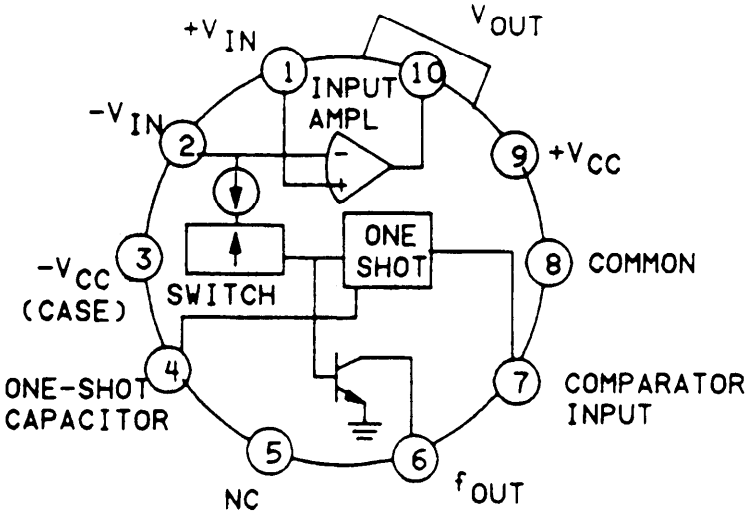
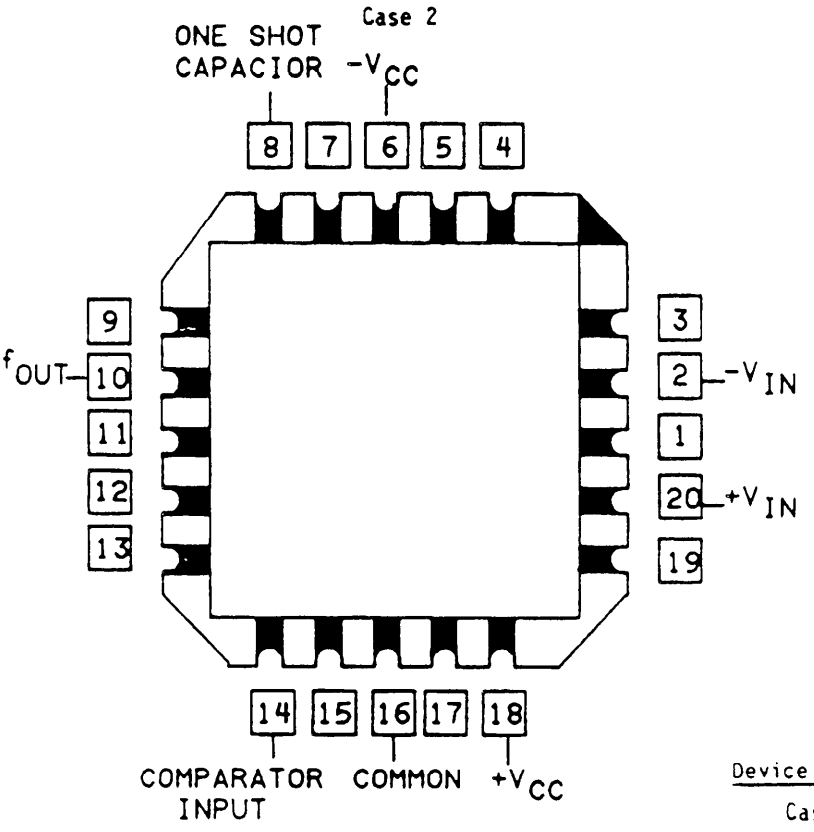


FIGURE 1. Circuit diagrams and terminal connections.

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Device type 02

Case 2



Device type 03

Case C

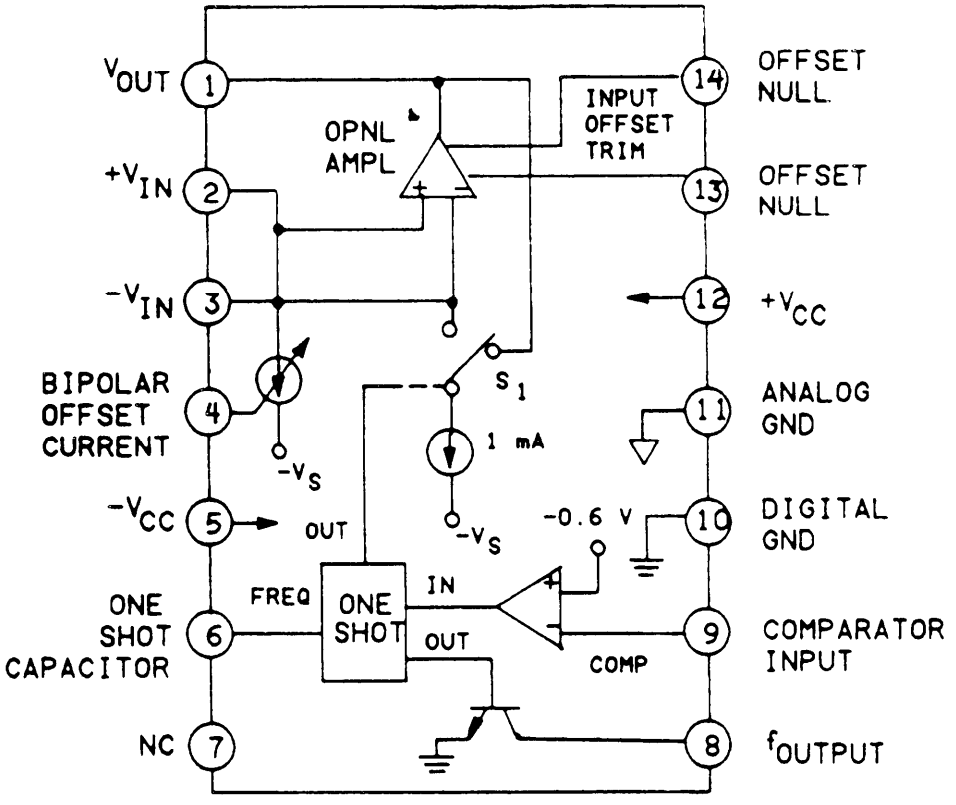


FIGURE 1. Circuit diagrams and terminal connections - Continued.

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Device type 01

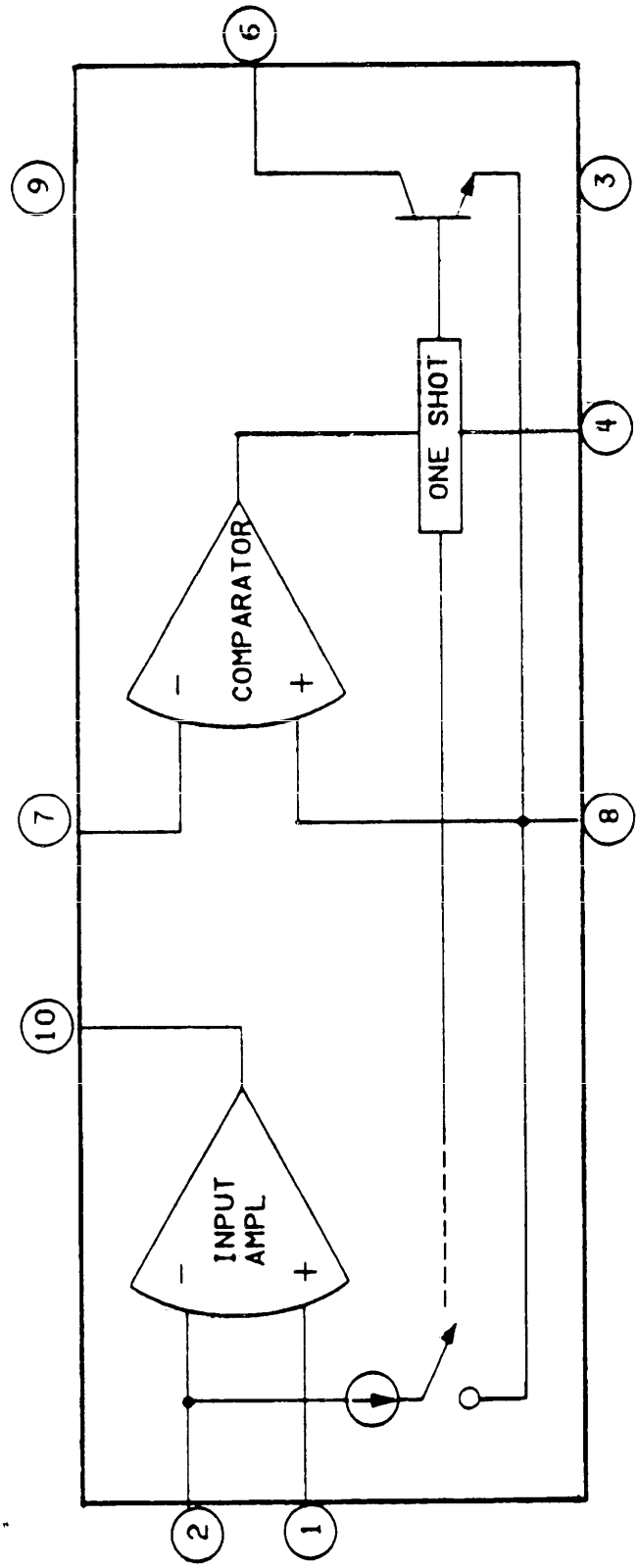


FIGURE 2. Funtional block diagrams.

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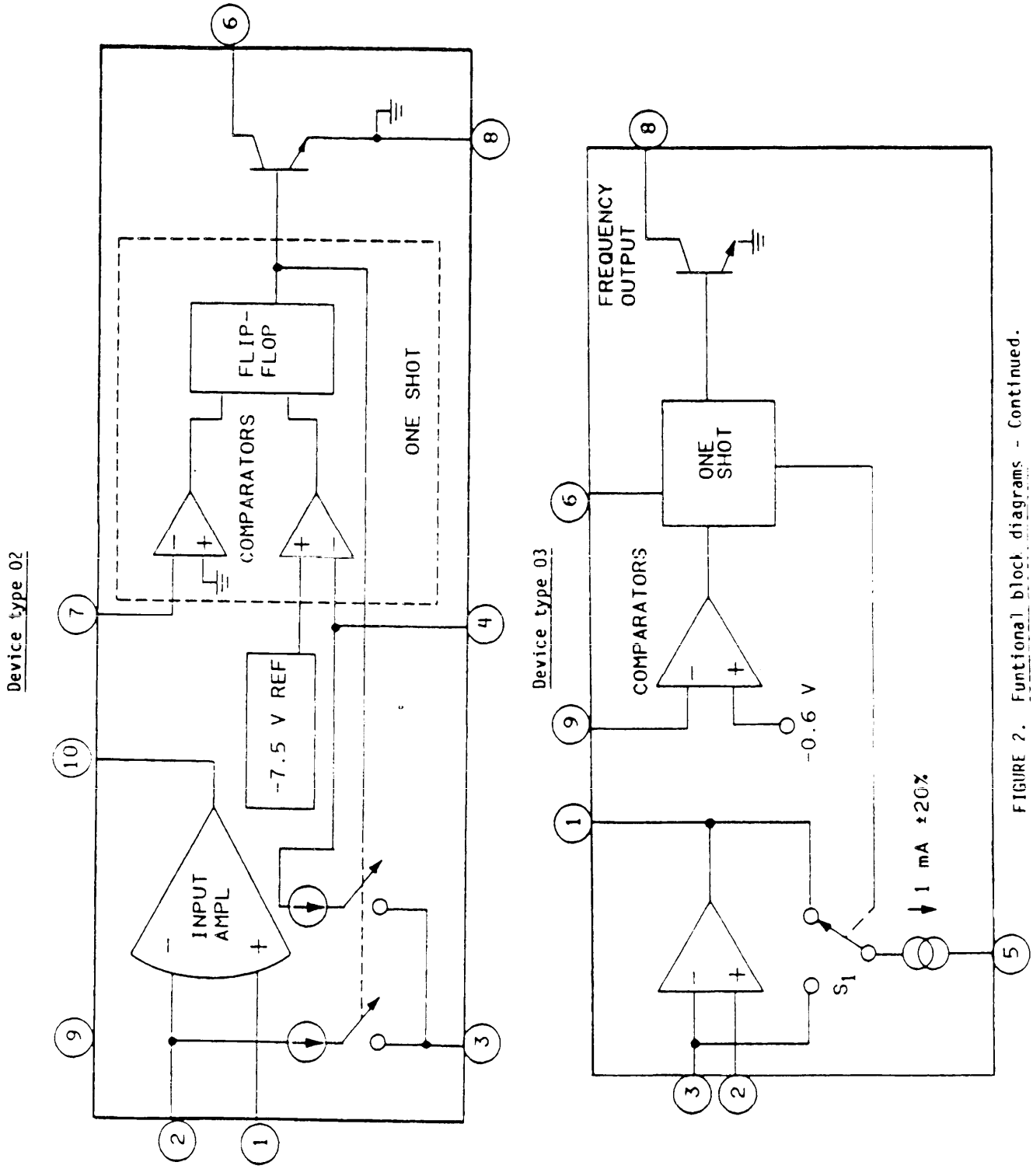
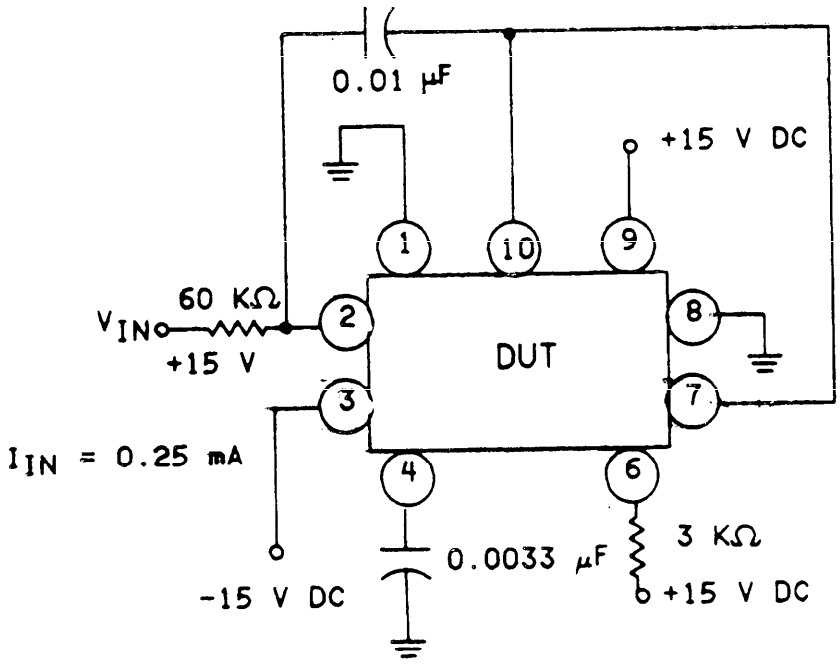


FIGURE 2. Funtional block diagrams - Continued.

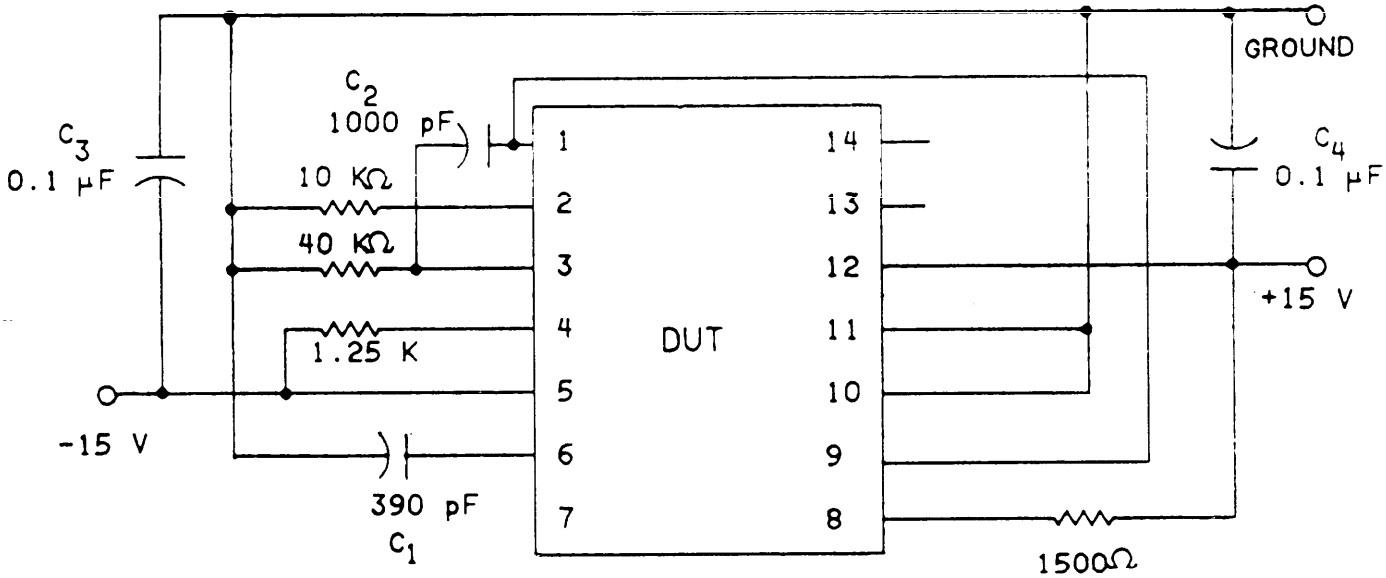
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Device types 01 and 02



NOTES:
1. Capacitors are mica, 100 V, ±5%.
2. Resistors are metal film, 1/4 W, 1%.

Device type 03



NOTES:
1. Capacitors C1 and C2 mica, 100 V, ±5%.
2. Resistors are RN55C 1%, metal film 1/4 W.
3. Capacitors C3 and C4 ceramic, 50 V, ±20%.

FIGURE 3. Test circuit for burn-in and steady-state life test.

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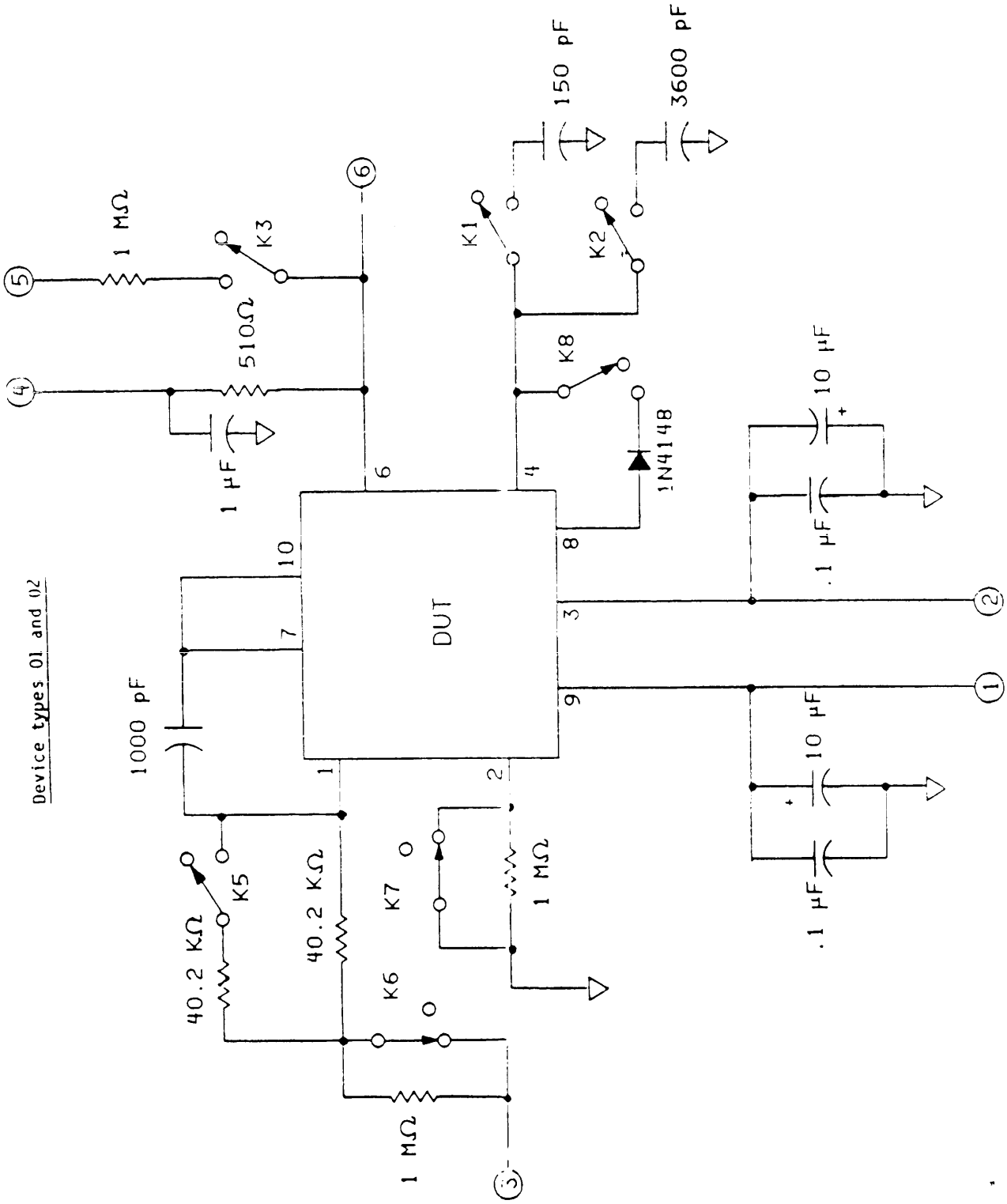
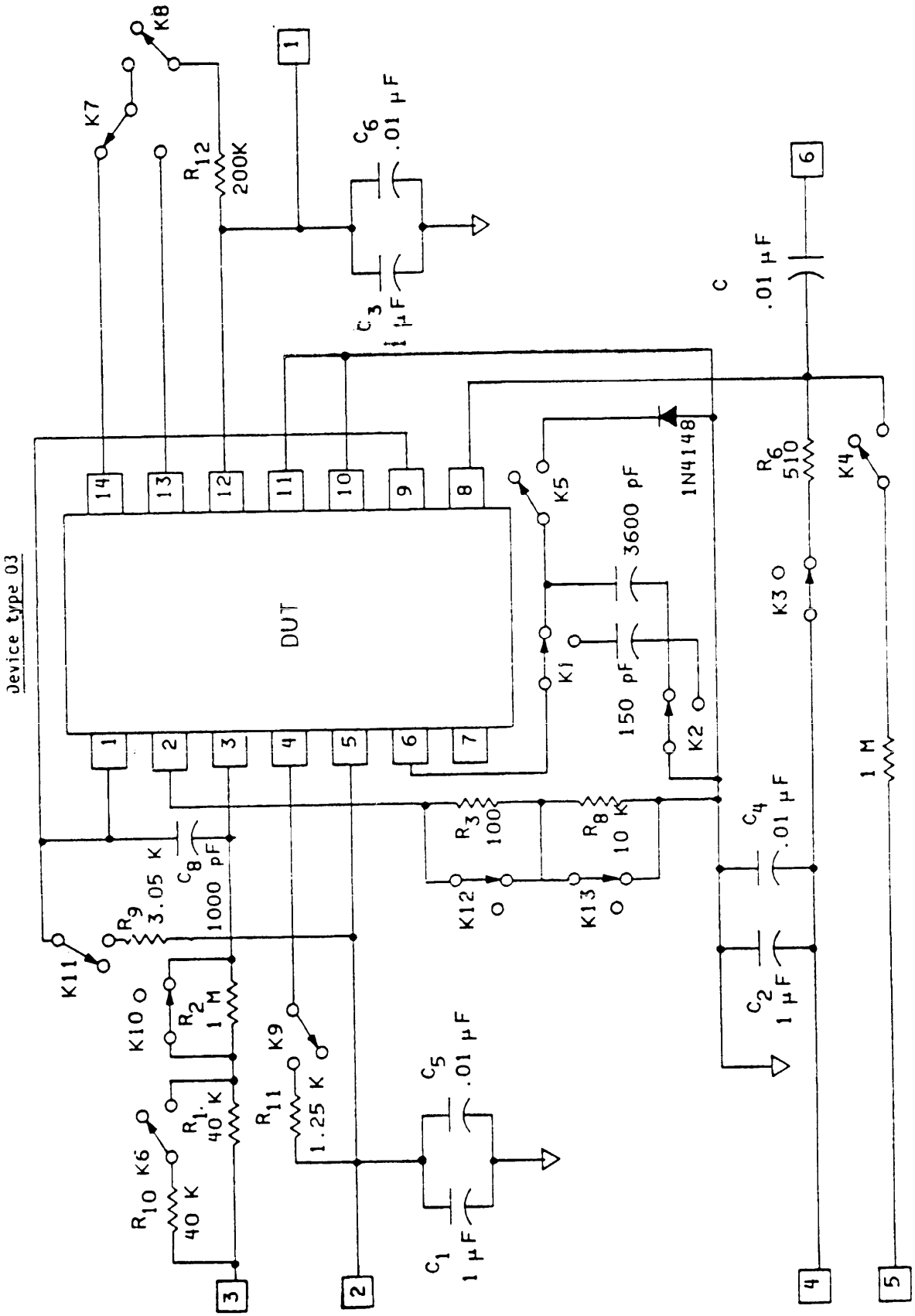


FIGURE 4. Test circuit for ac test conditions.

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NOTE: Resistance values are in ohms.

FIGURE 4. Test circuit for ac test conditions - Continued.

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Device type 03

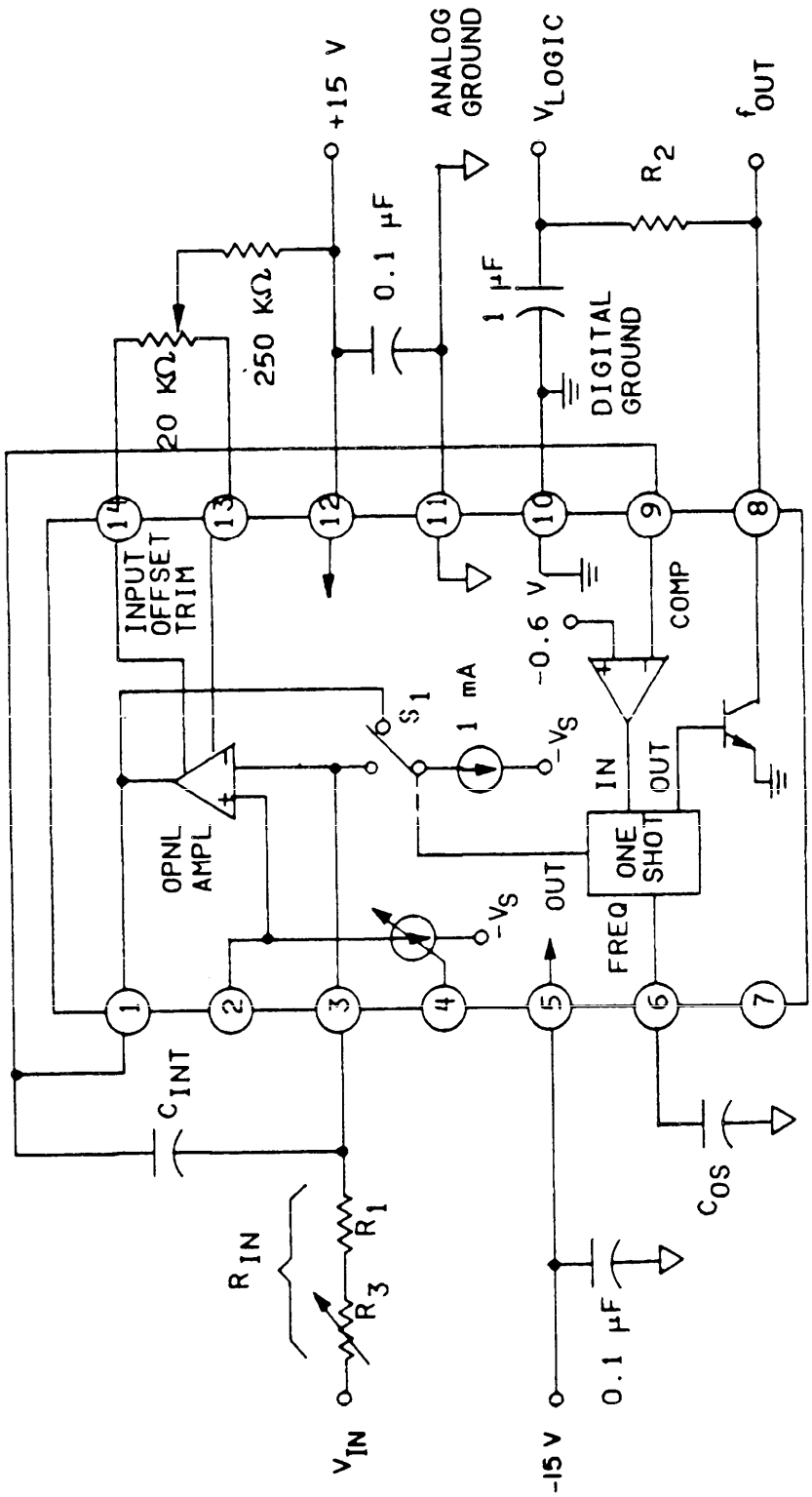


FIGURE 5. Voltage-to-frequency circuit (positive input) - Continued.

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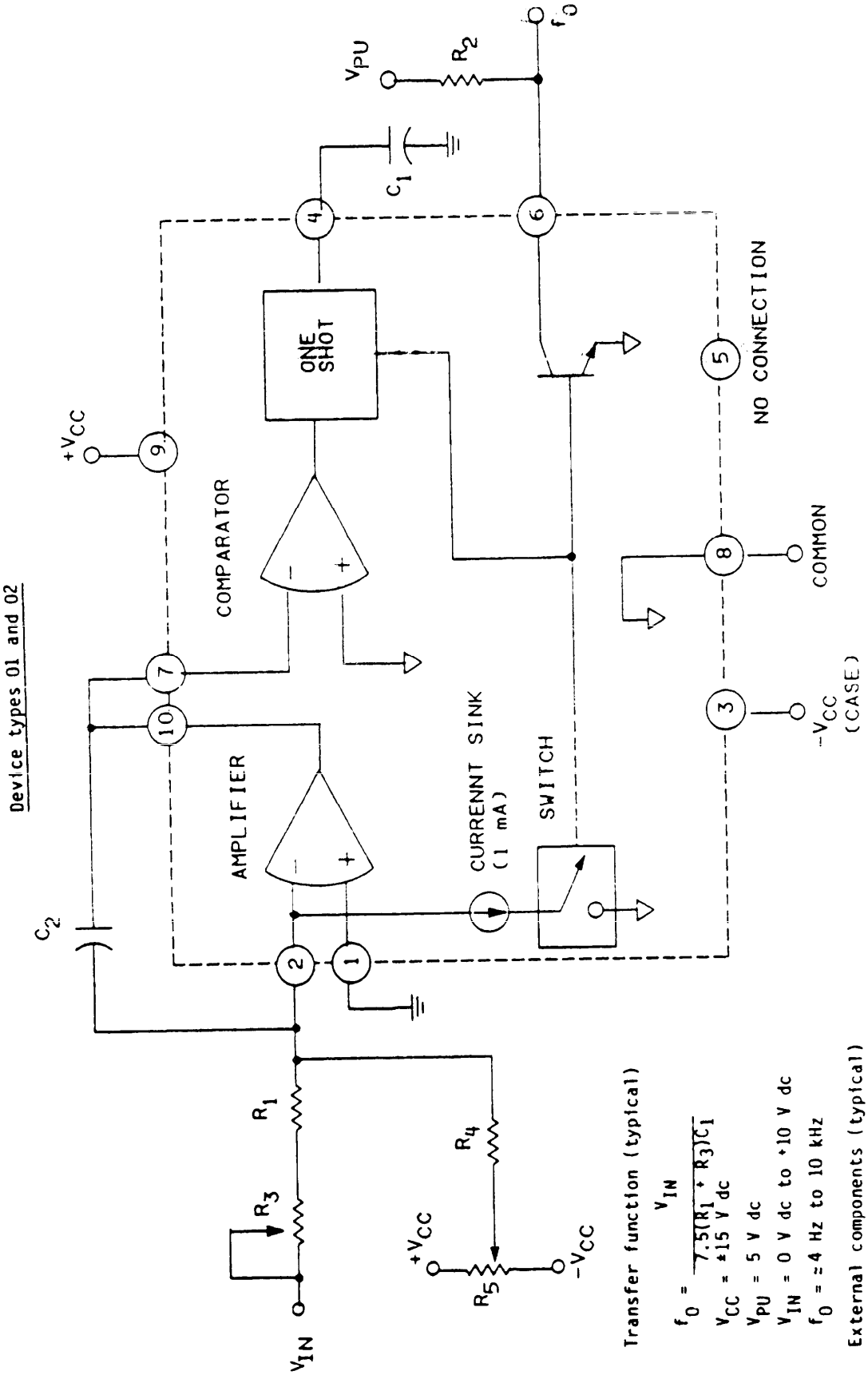
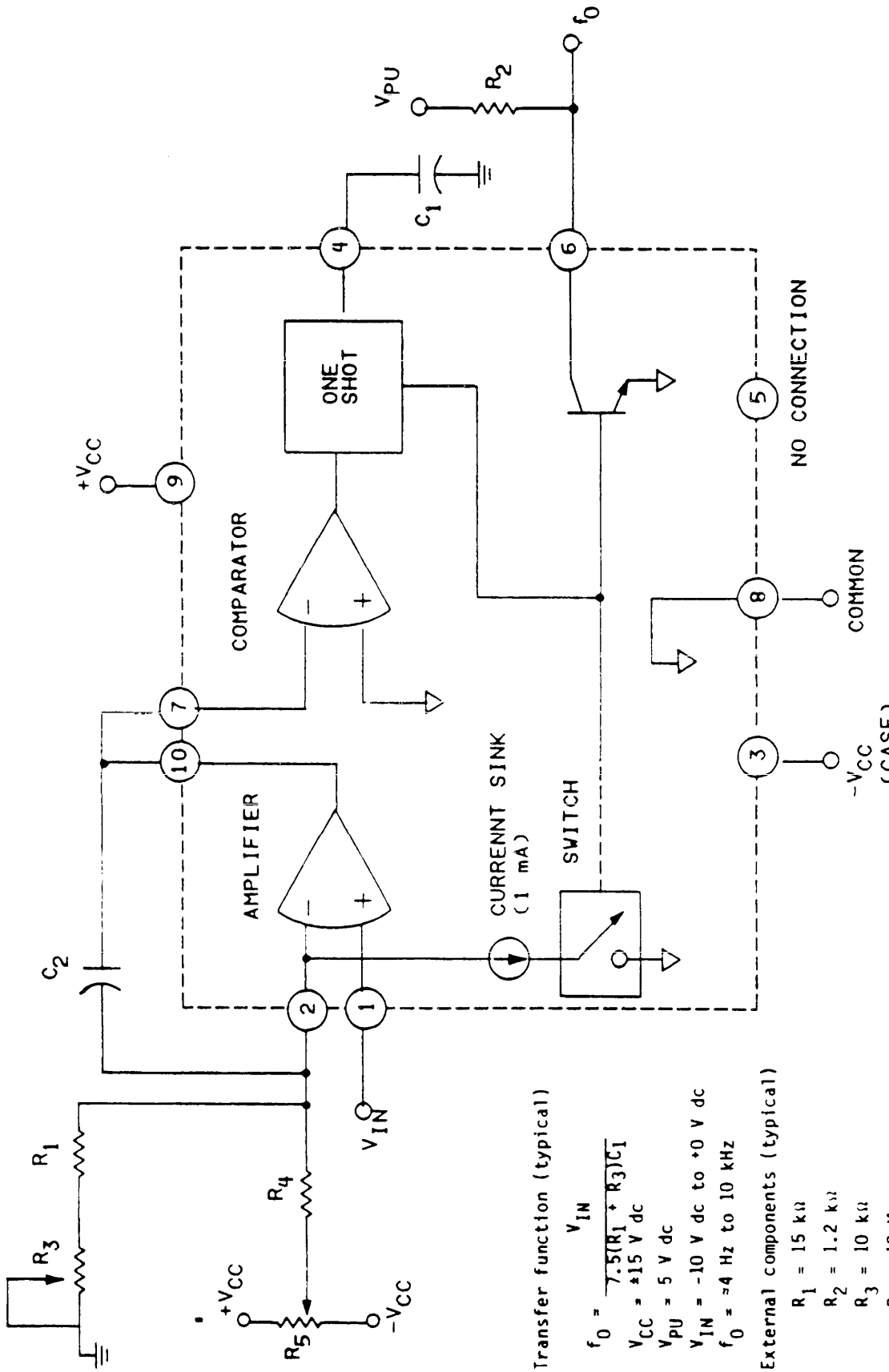


FIGURE 5. Voltage-to-frequency circuit (positive input).

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Device types 01 and 02



Transfer function (typical)

$$f_O = \frac{V_{IN}}{7.5(R_1 + R_3)C_1}$$

V_{CC} = ±15 V dc
V_{PU} = 5 V dc
V_{IN} = -10 V dc to +0 V dc
f_O = 4 Hz to 10 kHz

External components (typical)

R_1 = 15 kΩ
R_2 = 1.2 kΩ
R_3 = 10 kΩ
R_4 = 10 MΩ
R_5 = 50 kΩ
C_1 = 6800 pF (0 = 0.5)
C_2 = 0.01 µF

FIGURE 6. Voltage-to-frequency circuit (negative input).

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Device type 03

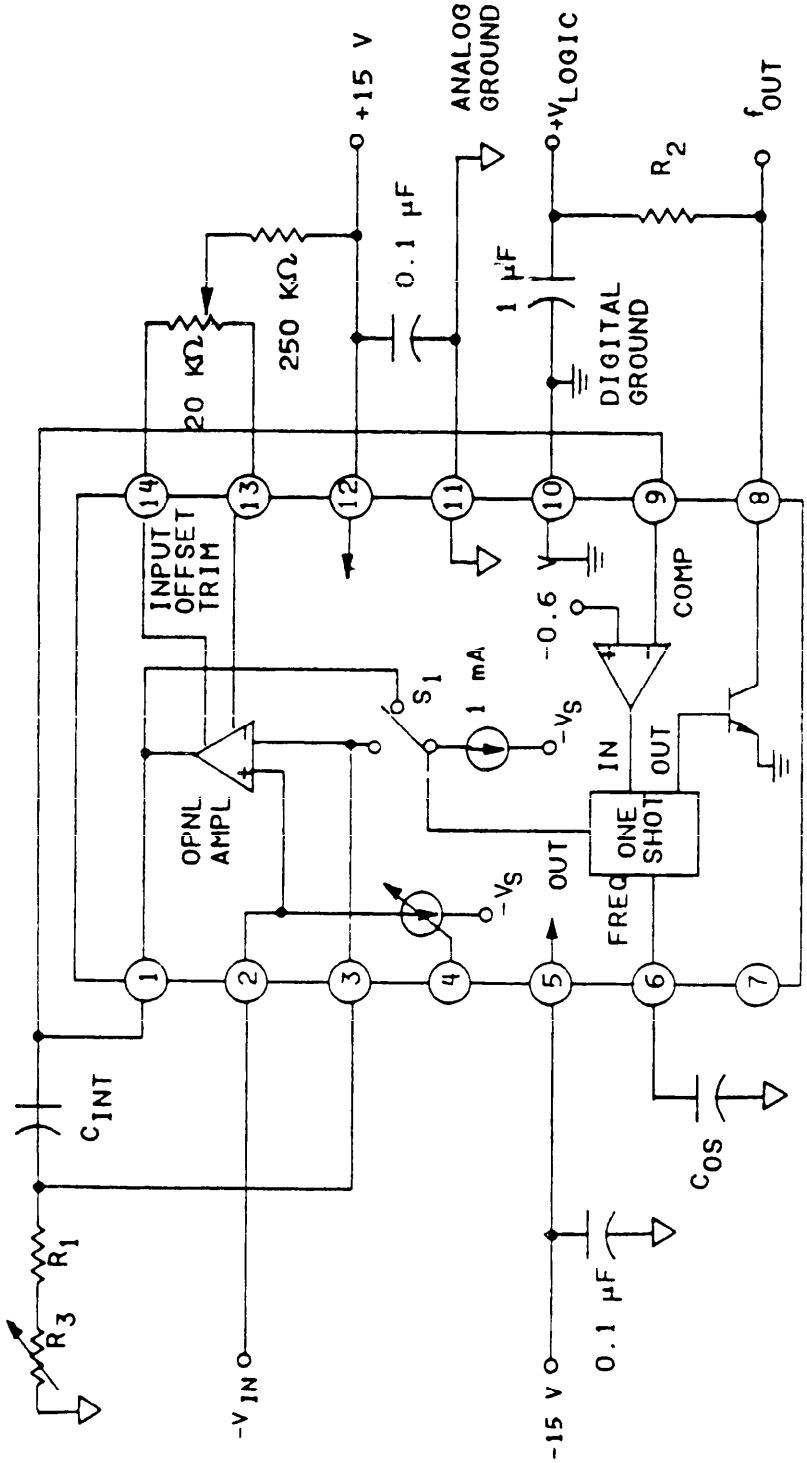


FIGURE 6. Voltage-to-frequency circuit (negative input) - Continued.

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Device type 03

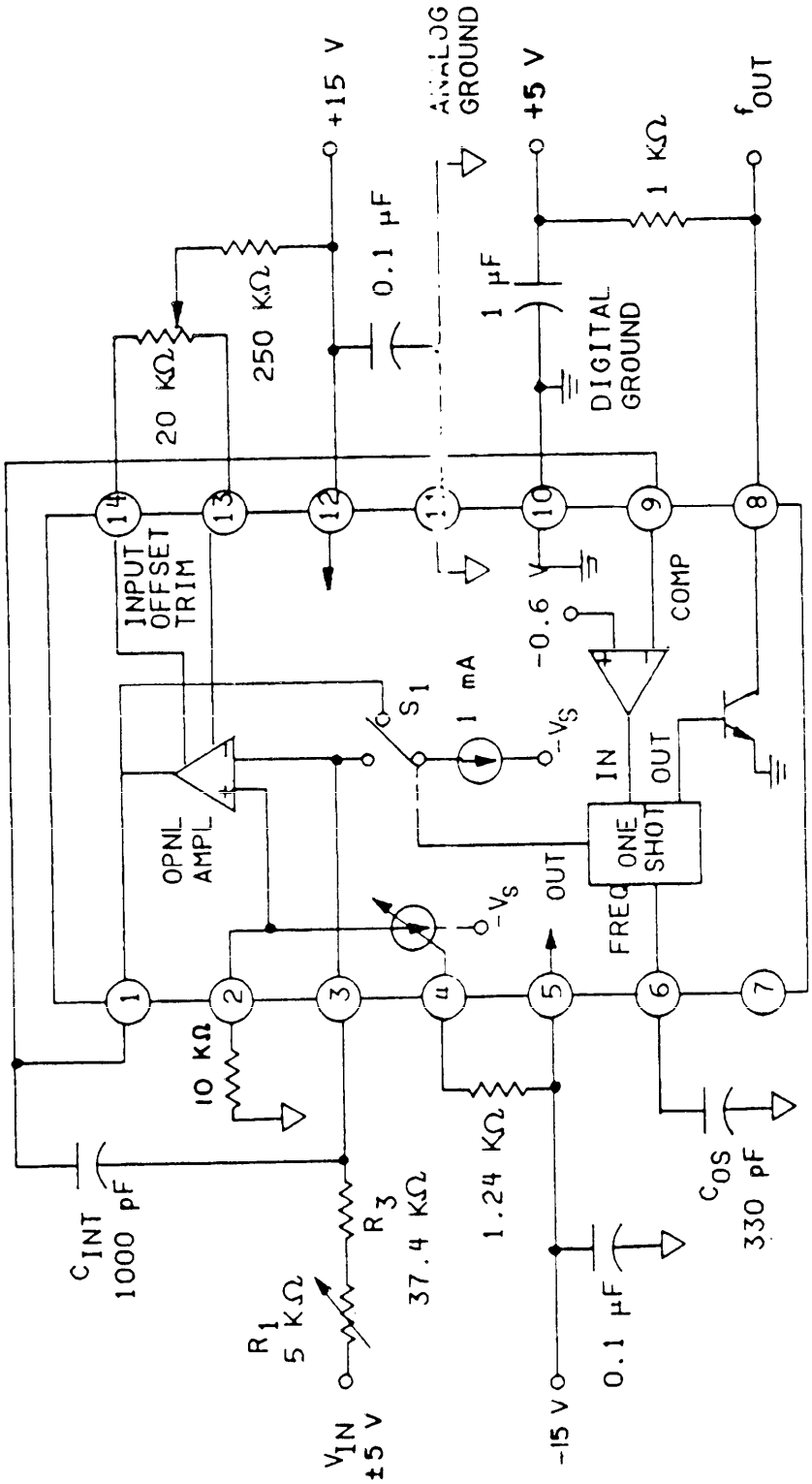


FIGURE 7. Voltage-to-frequency circuit (bipolar).

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TABLE III. Group A inspection for device type 01.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Notes	Applied voltages and currents ref. pin numbers (see figure 4) 1	2	3	4	5	Energized relays	Measured pin No.	Value	Unit	Equation	Limits Min	Max	Unit
1 T_A = +25°C	V_{OL1}		1		15 V	-15 V	0 V	8 mA		K2	6	E1	V	V_{OL1} = E1	0	0.4	V
	V_{OL2}		2		15 V	15 V	0 V	16 mA		"	6	E2	V	V_{OL2} = E2	0	0.8	
	I_{OH}		3						17 V	K3	5	I1	mA	I_{OH} = I1	---	1	µA
	I_{CC}		4		15 V	-15 V	0 V			None	1	I2	"	I_{CC} = I2	0	8.0	mA
	I_{EE}		5		"	"	0 V			None	2	I3	"	I_{EE} = I3	-8.0	0	
	L_E		6	f = 10K	"	"	.01 V	5 V		K2	6	F1	Hz	L_E = .01(F(N)-(F12*(V_{IN}-V_{OS1}))/(10-V_{OS1})	-.01	.01	
			7	"	"	"	.1 V	"		"	"	F2	"		"	"	
			8	"	"	"	1 V	"		"	"	F3	"		"	"	
			9	"	"	"	2 V	"		"	"	F4	"	N = 1 to 12	"	"	
			10	"	"	"	3 V	"		"	"	F5	"		"	"	
			11	"	"	"	4 V	"		"	"	F6	"		"	"	
			12	"	"	"	5 V	"		"	"	F7	"		"	"	
			13	"	"	"	6 V	"		"	"	F8	"		"	"	
			14	"	"	"	7 V	"		"	"	F9	"		"	"	
			15	"	"	"	8 V	"		"	"	F10	"		"	"	
			16	"	"	"	9 V	"		"	"	F11	"		"	"	
			17	"	"	"	10 V	"		"	"	F12	"		"	"	% of FSR
			18	f = 200K	"	"	.01 V	"		K1,K5	"	F13	Hz	L_E = .0005(F(N)-(F24*(V_{IN}-V_{OS2}))/(10-V_{OS2})	-.1	.1	
			19	"	"	"	.1 V	"		"	"	F14	"		"	"	
			20	"	"	"	1 V	"		"	"	F15	"		"	"	
			21	"	"	"	2 V	"		"	"	F16	"	N = 13 to 24	"	"	
			22	"	"	"	3 V	"		"	"	F17	"		"	"	
			23	"	"	"	4 V	"		"	"	F18	"		"	"	
			24	"	"	"	5 V	"		"	"	F19	"		"	"	
			25	"	"	"	6 V	"		"	"	F20	"		"	"	
			26	"	"	"	7 V	"		"	"	F21	"		"	"	
			27	"	"	"	8 V	"		"	"	F22	"		"	"	
			28	"	"	"	9 V	"		"	"	F23	"		"	"	
			29	"	"	"	10 V	"		"	"	F24	"		"	"	
	V_{OS1}		30	f = 10K	"	"	.01 V	"		K2	"	F1	"	V_{OS1} = .01-F1*9.99/(F12-F1)	-4	4	mV
				f = 10K	"	"	10 V	"		K2	"	F12	"				
	V_{OS2}		31	f = 200K	"	"	.01 V	"		K1	"	F13	"	V_{OS2} = .01-F13*9.99/(F24-F13)	-4	4	mV
				f = 200K	"	"	10 V	"		K5	"	F24	"				
	$+I_{IB}$	4001	32	f = 10K	"	"	.2 V	"		K2	"	F25	"	V_{OS3} = .2-F25*9.8/(F26-F25) $+I_{IB}$=(V_{OS3}-V_{OS1})/10K	-150	150	nA
				"	"	"	10 V	"		K7	"	F26	"				
	$-I_{IB}$	4001	33	"	"	"	.1 V	"		K2	"	F27	"	V_{OS4} = .1-F27*9.9/(F28-F27) $+I_{IB}$=(V_{OS4}-V_{OS1})/10K	-90	90	nA
				"	"	"	10 V	"		K6	"	F28	"				
	A_E		34	"	"	"	"	"		K2	"	F12	"	A_{E1} = .01((F12-F1)x10/9.99-10K)	-10	10	% of FSR
			35	f = 200K	"	"	"	"		K1,K5	"	F24	"	A_{E2} = .0005((F24-F13)x10/9.99-200K)	-10	10	% of FSR
	P_{SS1}		36	f = 10K	11 V	"	"	"		K2	"	F29	"	P_{SS1} = 2.5(F29-F30)/F11	-.04	.04	% of FSR /% supply
			37	f = 10K	17 V	"	"	"		K2	"	F30	"				
			38	f = 200K	11 V	"	"	"		K1,K5	"	F31	"	P_{SS1} = 2.5(F31-F32)/F22	-.04	.04	
			39	f = 200K	17 V	"	"	"		K1,K5	"	F32	"				

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TABLE II. Group A inspection - Device type 01 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Notes	Applied voltages and currents ref. pin numbers (see figure 4) 1	2	3	4	5	Energized relays	Measured pin No.	Measured pin Value	Measured pin Unit	Equation	Limits Min	Limits Max	Limits Unit
1 T_A = +25°C	P_{SS2}		40 41	f = 10K f = 10K	15 V "	-11 V -17 V	10 V "	5 V "		K2 K2	6 "	F33 F34	Hz "	P_{SS2} = 2.5(F33-F34)/F11	-.04	.04	% of FSR /% supply
			42 43	f = 200K f = 200K	" "	-11 V -17 V	" "	" "		K1,K5 K1,K5	" "	F35 F36	" "	P_{SS2} = 2.5(F35-F36)/F22	-.04	.04	
	F_{SE}		44 45	f = 10K f = 200K	" "	-15 V -15 V	" "	" "		K2 K1,K5	" "	F12 F24	" "	F_{SE1} = F12 F_{SE2} = F24	-10 -10	10 10	% of FSR
2 T_A = +125°C	V_{OL1}		46		15 V	-15 V	0 V	5 mA		K2	6	E3	V	V_{OL1} = E3	0	0.4	V
	I_{CC}	3005	47		"	"	"			None	1	I4	mA	I_{CC} = I4	0	8.0	mA
	I_{EE}	3005	48		"	"	"	"		None	2	I5	mA	I_{EE} = I5	-8.0	0	mA
	L_E		49 50 51 52 53 54 55 56 57 58 59 60	f = 10K " " " " " " " " " " "	" " " " " " " " " " " "	" " " " " " " " " " " "	.01 V .1 V 1 V 2 V 3 V 4 V 5 V 6 V 7 V 8 V 9 V 10 V	5 V " " " " " " " " " " "		K2 " " " " " " " " " " "	6 " " " " " " " " " " "	F37 F38 F39 F40 F41 F42 F43 F44 F45 F46 F47 F48	Hz " " " " " " " " " " "	L_E = .01(F(N)-(F48*(V_{IN}-V_{OS1})) /(10-V_{OS1}) N = 37 to 48	-.01 " " " " " " " " " " "	.01 " " " " " " " " " " "	% of FSR
			61 62 63 64 65 66 67 68 69 70 71 72	f = 200K " " " " " " " " " " "	" " " " " " " " " " " "	" " " " " " " " " " " "	.01 V .1 V 1 V 2 V 3 V 4 V 5 V 6 V 7 V 8 V 9 V 10 V	" " " " " " " " " " " "		K1,K5 " " " " " " " " " " "	" " " " " " " " " " " "	F49 F50 F51 F52 F53 F54 F55 F56 F57 F58 F59 F60	" " " " " " " " " " " "	L_E = .0005(F(N)-(F60*(V_{IN}-V_{OS2})) /(10-V_{OS2}) N = 49 to 60	-.1 " " " " " " " " " " "	.1 " " " " " " " " " " "	
	V_{OS1}		73	f = 10K f = 10K	" "	" "	.01 V 10 V	" "		K2 K2	" "	F37 F48	" "	V_{OS1} = .01-F37*9.99/(F48-F37)	-7.0	7.0	mV
	V_{OS2}		74	f = 200K f = 200K	" "	" "	.01 V 10 V	" "		K1,K5 K1,K5	" "	F49 F60	" "	V_{OS2} = .01-F49*9.99/(F60-F49)	-7.0	7.0	
	A_E		75	f = 10K	"	"	"	"		K2	"	F48	"	A_{E1} = .01((F48-F37)x10/9.99-10K)	-10	10	% FSR
			76	f = 200K	"	"	"	"		K1,K5	"	F60	"	A_{E2} = .0005((F60-F49)x10/9.99-200K)	-10	10	
	dV_{OS}/dT		77	f = 10K										dV_{OS}/dT = (V_{OS1}(+125°C)-V_{OS1}(+25°C))/100	-30	30	µV/°C
			78	f = 200K										dV_{OS}/dT = (V_{OS2}(+125°C)-V_{OS2}(+25°C))/100	-30	30	
	dA_E/dT		79	f = 10K										dA_{E1}/dT = (A_{E1}(+125°C)-A_{E1}(+25°C))/100	-200	200	ppm/°C
			80	f = 200K										dA_{E2}/dT = (A_{E2}(+125°C)-A_{E2}(+25°C)/100	-200	200	

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TABLE III. Group A inspection for device type 01 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Notes	Applied voltages and currents ref. pin numbers (see figure 4) 1	2	3	4	5	Energized relays	Measured pin No.	Value	Unit	Equation	Limits Min	Max	Unit
2 T_A = +125°C	$d_F(FS)/dt$		81	f = 10K							6	F48	Hz	$d_F(FS1)/dT = FS1(+125°C)-FS1(25°C)/100$	-120	200	ppm of FSR/°C
			82	f = 200K							6	F60	Hz	$d_F(FS2)dT = FS2(+125°C)-FS2(+25°C)/100$	-150	150	
3 T_A = -55°C	V_{OL1}		83		15 V	-15 V	0 V	5 mA		K2	6	E4	V	$V_{OL1} = E4$	0	0.4	V
	I_{CC}	3005	84		"	"	"			None	1	I6	mA	$I_{CC} = I6$	0	8.0	mA
	I_{EE}	3005	85		"	"	"			None	2	I7	mA	$I_{EE} = I7$	-8.0	0	mA
	L_E		86	f = 10K	"	"	.01 V	5 V		K2	6	F61	Hz	$L_E = .01(F(N)-(F72*(V_{IN}-V_{OS1}))/(10-V_{OS1})$	-.01	.01	% of FSR
			87	"	"	"	.1 V	"		"	"	F62	"		"	"	
			88	"	"	"	1 V	"		"	"	F63	"		"	"	
			89	"	"	"	2 V	"		"	"	F64	"	N = 61 to 72	"	"	
			90	"	"	"	3 V	"		"	"	F65	"		"	"	
			91	"	"	"	4 V	"		"	"	F66	"		"	"	
			92	"	"	"	5 V	"		"	"	F67	"		"	"	
			93	"	"	"	6 V	"		"	"	F68	"		"	"	
			94	"	"	"	7 V	"		"	"	F69	"		"	"	
			95	"	"	"	8 V	"		"	"	F70	"		"	"	
			96	"	"	"	9 V	"		"	"	F71	"		"	"	
			97	"	"	"	10 V	"		"	"	F72	"		"	"	
			98	f = 200K	"	"	.01 V	"		K1,K5	"	F73	"	$L_E = .0005(F(N)-(F84*(V_{IN}-V_{OS2}))/(10-V_{OS2})$	-0.1	0.1	
			99	"	"	"	.1 V	"		"	"	F74	"		"	"	
			100	"	"	"	1 V	"		"	"	F75	"		"	"	
			101	"	"	"	2 V	"		"	"	F76	"	N = 73 to 84	"	"	
			102	"	"	"	3 V	"		"	"	F77	"		"	"	
			103	"	"	"	4 V	"		"	"	F78	"		"	"	
			104	"	"	"	5 V	"		"	"	F79	"		"	"	
			105	"	"	"	6 V	"		"	"	F80	"		"	"	
			106	"	"	"	7 V	"		"	"	F81	"		"	"	
			107	"	"	"	8 V	"		"	"	F82	"		"	"	
			108	"	"	"	9 V	"		"	"	F83	"		"	"	
			109	"	"	"	10 V	"		"	"	F84	"		"	"	
	V_{OS1}		110	f = 10K f = 10K	"	"	.01 V 10 V	"		K2 K2	"	F61 F72	"	$V_{OS1} = .01-F61*9.99/(F72-F61)$	-7.0	7.0	mV
	V_{OS2}		111	f = 200K f = 200K	"	"	.01 V 10 V	"		K1 K5	"	F73 F84	"	$V_{OS2} = .01-F73*9.99/(F84-F73)$	-7.0	7.0	
	A_E		112	f = 10K	"	"	"	"		K2	"	F72	"	$A_{E1} = .01((F72-F61)*10/9.99-10K)$	-10	10	% of FSR
			113	f = 200K	"	"	"	"		K1,K5	"	F84	"	$A_{E2} = .0005((F84-F73)*10/9.99-200K)$	-10	10	
	$d_{V_{OS}}/dt$		114	f = 10K										$d_{V_{OS}}/dT = V_{OS1}(-55°C)-V_{OS1}(+25°C)/80$	-30	30	µV/°C
			115	f = 200K										$d_{V_{OS}}/dT = V_{OS2}(-55°C)-V_{OS2}(+25°C)/80$	-30	30	
	d_{A_E}/dT		116	f = 10K										$d_{AE1}/dT = A_{E1}(-55°C)-A_{E1}(+25°C)/80$	-200	200	ppm/°C
			117	f = 200K										$d_{AE2}/dT = A_{E2}(-55°C)-A_{E2}(+25°C)/80$	-200	200	
	$d_F(FS)/dt$		118	f = 10K							6	F72	Hz	$d_F(FS1)/dT = FS1(-55°C)-FS1(+25°C)/80$	-200	120	ppm of FSR/°C
			119	f = 200K							6	F84	Hz	$d_F(FS2)/dT = FS2(-55°C)-FS2(+25°C)/80$	-400	100	

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TABLE III. Group A Inspe or device type 02.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Notes	Applied voltages and currents ref. pin numbers (see figure 4) 1	2	3	4	5	Energized relays	Measured pin No.	Value	Unit	Equation	Limits Min	Max	Unit
1 T_A = +25°C	V_{OL1}		1		15 V	-15 V	0 V	8 mA		K2	6	E1	V	V_{OL1} = E1	0	0.4	V
	V_{OL2}		2		15 V	-15 V	0 V	5 mA		-	6	E2	V	V_{OL2} = E2	0	0.8	
	I_{OH}		3						17 V	K3	5	I1	mA	I_{OH} = I1		1	µA
	I_{CC}		4		15 V	-15 V	0 V			None	1	I2	"	I_{CC} = I2	0	8.0	mA
	I_{EE}		5		"	"	0 V			None	2	I3	"	I_{EE} = I3	-8.0	0	
	L_E		6	f = 10K	"	"	.01 V	5 V		K2	6	F1	Hz	L_E = .01(F(N)-(F12*(V_{IN}-V_{OS1}))/(10-V_{OS1})	-.005	.005	
			7	"	"	"	.1 V	"		"	"	F2	"		"	"	
			8	"	"	"	1 V	"		"	"	F3	"		"	"	
			9	"	"	"	2 V	"		"	"	F4	"	N = 1 to 12	"	"	
			10	"	"	"	3 V	"		"	"	F5	"		"	"	
			11	"	"	"	4 V	"		"	"	F6	"		"	"	
			12	"	"	"	5 V	"		"	"	F7	"		"	"	
			13	"	"	"	6 V	"		"	"	F8	"		"	"	
			14	"	"	"	7 V	"		"	"	F9	"		"	"	
			15	"	"	"	8 V	"		"	"	F10	"		"	"	
			16	"	"	"	9 V	"		"	"	F11	"		"	"	
			17	"	"	"	10 V	"		"	"	F12	"		"	"	% of FSR
			18	f = 1M	"	"	.01 V	"		K1,K5	"	F13	"	L_E = .0005(F(N)-(F24*(V_{IN}-V_{OS2}))/(10-V_{OS2})	-0.1	0.1	
			19	"	"	"	.1 V	"		"	"	F14	"		"	"	
			20	"	"	"	1 V	"		"	"	F15	"		"	"	
			21	"	"	"	2 V	"		"	"	F16	"	N = 13 to 24	"	"	
			22	"	"	"	3 V	"		"	"	F17	"		"	"	
			23	"	"	"	4 V	"		"	"	F18	"		"	"	
			24	"	"	"	5 V	"		"	"	F19	"		"	"	
			25	"	"	"	6 V	"		"	"	F20	"		"	"	
			26	"	"	"	7 V	"		"	"	F21	"		"	"	
			27	"	"	"	8 V	"		"	"	F22	"		"	"	
			28	"	"	"	9 V	"		"	"	F23	"		"	"	
			29	"	"	"	10 V	"		"	"	F24	"		"	"	
	V_{OS1}		30	f = 10K	"	"	.01 V	"		K2	"	F1	"	V_{OS1} = .01-F1*9.99/(F12-F1)	-0.5	0.5	mV
				f = 10K	"	"	10 V	"		K2	"	F12	"				
	V_{OS2}		31	f = 1M	"	"	.01 V	"		K1	"	F13	"	V_{OS2} = .01-F13*9.99/(F24-F13)	-0.5	0.5	mV
				f = 1M	"	"	10 V	"		K5	"	F24	"				
	$+I_{IB}$	4001	32	f = 10K	"	"	.2 V	"		K2	"	F25	"	V_{OS3} = .2-F25*9.8/(F26-F25)	-30	30	nA
				"	"	"	10 V	"		K7	"	F26	"	$+I_{IB}$=(V_{OS3}-V_{OS1})/10K			
	$-I_{IB}$	4001	33	"	"	"	.1 V	"		K2	"	F27	"	V_{OS4} = .1-F27*9.9/(F28-F27)	8	8	nA
				"	"	"	10 V	"		K6	"	F28	"	$+I_{IB}$=(V_{OS4}-V_{OS1})/10K			
	A_E		34	"	"	"	"	"		K2	"	F12	"	A_{E1} = .01((F12-F1)x10/9.99-10K)	-10	10	% of FSR
			35	f = 1M	"	"	"	"		K1,K5	"	F24	"	A_{E2} = .0001((F24-F13)x10/9.99-1M)	-10	10	% of FSR
	P_{SS1}		36	f = 10K	11 V	"	"	"		K2	"	F29	"	P_{SS1} = 2.5(F29-F30)/F11	-.025	.025	% of FSR 1/% supply
			37	f = 10K	17 V	"	"	"		K2	"	F30	"				
			38	f = 1M	11 V	"	"	"		K1,K5	"	F31	"	P_{SS1} = 2.5(F31-F32)/F22	-.025	.025	
			39	f = 1M	17 V	"	"	"		K1,K5	"	F32	"				

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TABLE III. Group A inspection for device type 02 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Notes	Applied voltages and currents ref. pin numbers (see figure 4) 1	2	3	4	5	Energized relays	Measured pin No.	Value	Unit	Equation	Limits Min	Max	Unit
1 T_A = +25°C	P_{SS2}		40	f = 10K	15 V	-11 V	10 V	5 V		K2	6	F33	Hz	P_{SS2} = 2.5(F33-F34)/F11	-.025	.025	% of FSR
			41	f = 10K	"	-17 V	"	"		K2	"	F34	"				/% supply
			42	f = 1M	"	-11 V	"	"		K1,K5	"	F35	"	P_{SS2} = 2.5(F35-F36)/F22	"	"	
			43	f = 1M	"	-17 V	"	"		K1,K5	"	F36	"				
	F_{SE}		44	f = 10K	"	-15 V	"	"		K2	"	F12	"	F_{SE1} = F12	"	"	% of FSR
			45	f = 1M	"	"	"	"		K1,K5	"	F24	"	F_{SE2} = F24			/% supply
2 T_A = +125°C	V_{OL1}		46		15 V	-15 V	0 V	5 mA		K2	6	E3	V	V_{OL1} = E3	0	0.4	V
	I_{CC}	3005	47		"	"	"			None	1	I4	mA	I_{CC} = I4	0	6.7	mA
	I_{EE}	3005	48		"	"	"	"		None	2	I5	mA	I_{EE} = I5	-6.7	0	mA
	L_E		49	f = 10K	"	"	.01 V	5 V		K2	6	F37	Hz	L_E = .01(F(N)-(F48*(V_{IN}-V_{OS1}))/(10-V_{OS1})	-.005	.005	% of FSR
			50		"	"	.1 V	"		"	"	F38	"		"	"	
			51		"	"	1 V	"		"	"	F39	"		"	"	
			52		"	"	2 V	"		"	"	F40	"	N = 37 to 48	"	"	
			53		"	"	3 V	"		"	"	F41	"		"	"	
			54		"	"	4 V	"		"	"	F42	"		"	"	
			55		"	"	5 V	"		"	"	F43	"		"	"	
			56		"	"	6 V	"		"	"	F44	"		"	"	
			57		"	"	7 V	"		"	"	F45	"		"	"	
			58		"	"	8 V	"		"	"	F46	"		"	"	
			59		"	"	9 V	"		"	"	F47	"		"	"	
			60		"	"	10 V	"		"	"	F48	"		"	"	
			61	f = 1M	"	"	.01 V	"		K1,K5	"	F49	"	L_E = .0005(F(N)-(F60*(V_{IN}-V_{OS2}))/(10-V_{OS2})	-.1	.1	
			62		"	"	.1 V	"		"	"	F50	"		"	"	
			63		"	"	1 V	"		"	"	F51	"		"	"	
			64		"	"	2 V	"		"	"	F52	"	N = 49 to 60	"	"	
			65		"	"	3 V	"		"	"	F53	"		"	"	
			66		"	"	4 V	"		"	"	F54	"		"	"	
			67		"	"	5 V	"		"	"	F55	"		"	"	
			68		"	"	6 V	"		"	"	F56	"		"	"	
			69		"	"	7 V	"		"	"	F57	"		"	"	
			70		"	"	8 V	"		"	"	F58	"		"	"	
			71		"	"	9 V	"		"	"	F59	"		"	"	
			72		"	"	10 V	"		"	"	F60	"		"	"	
	V_{OS1}		73	f = 10K	"	"	.01 V	"		K2	"	F37	"	V_{OS1} = .01-F37*9.99/(F48-F37)	-0.6	0.6	mV
					"	"	10 V	"		K2	"	F48	"				
	V_{OS2}		74	f = 1M	"	"	.01 V	"		K1,K5	"	F49	"	V_{OS2} = .01-F49*9.99/(F60-F49)	-0.6	0.6	
					"	"	10 V	"		K1,K5	"	F60	"				
	A_E		75	f = 10K	"	"	10 V	"		K2	"	F48	"	A_{E1} = .01((F48-F37)x10/9.99-10K)	-10	10	% of FSR
			76	f = 1M	"	"	"	"		K1,K5	"	F60	"	A_{E2} = .0001((F60-F49)x10/9.99-1M)	-10	10	
	d_{VOS}/dT		77	f = 10K										d_{VOS}/dT = (V_{OS1}(+125°C)-V_{OS1}(+25°C))/100	-5	5	µV/°C
			78	f = 1M										d_{VOS}/dT = (V_{OS2}(+125°C)-V_{OS2}(+25°C))/100	-5	5	
	d_{AE}/dT		79	f = 10K										d_{AE1}/dT = A_{E1}(+125°C)-A_{E1}(+25°C)/100	-50	50	ppm/°C
			80	f = 1M										d_{AE2}/dT = A_{E2}(+125°C)-A_{E2}(+25°C)/100	-50	50	

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TABLE III. Group A inspection for device type 02 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Notes	Applied voltages and currents ref. pin numbers (see figure 4) 1	2	3	4	5	Energized relays	Measured pin No.	Value	Unit	Equation	Limits Min	Max	Unit
2 T_A = +125°C	$d_{F(FS)}/dT$		81	f = 10K							6	F48	Hz	$d_F(FS1)/dT$ = FS1(125°C) - FS1(25°C)/100	-50	50	ppm of FSR/°C
			82	f = 1M							6	F6J	Hz	$d_F(FS2)dT$ = FS2(125°C)-FS2(25°C)/100	-100	0	
3 T_A = -55°C	V_{OL1}		83		15 V	15 V	0 V	5 mA		K2	6	E4	V	V_{OL1} = E4	0	0.4	V
	I_{CC}	3005	84		"	"	"			None	1	I6	mA	I_{CC} = I6	0	8.0	mA
	I_{EE}	3005	85		"	"	"	"		None	2	I7	mA	I_{EE} = I7	-8.0	0	mA
	L_E		86	f = 10K	"	"	.01 V	5 V		K2	6	F61	Hz	L_E = .01(F(N)-(F72*(V_{IN}-V_{OS1}))	-.005	.005	
			87	"	"	"	.1 V	"		"	"	F62	"	/(10-V_{OS1})	"	"	
			88	"	"	"	1 V	"		"	"	F63	"		"	"	
			89	"	"	"	2 V	"		"	"	F64	"	N = 61 to 72	"	"	
			90	"	"	"	3 V	"		"	"	F65	"		"	"	
			91	"	"	"	4 V	"		"	"	F66	"		"	"	
			92	"	"	"	5 V	"		"	"	F67	"		"	"	
			93	"	"	"	6 V	"		"	"	F68	"		"	"	
			94	"	"	"	7 V	"		"	"	F69	"		"	"	
			95	"	"	"	8 V	"		"	"	F70	"		"	"	
			96	"	"	"	9 V	"		"	"	F71	"		"	"	
			97	"	"	"	10 V	"		"	"	F72	"		"	"	% of FSR
			98	f = 1M	"	"	.01 V	"		K1,K5	"	F73	"	L_E = .0005(F(N)-(F84*(V_{IN}-V_{OS2}))	-0.1	0.1	
			99	"	"	"	.1 V	"		"	"	F74	"	/(10-V_{OS2})	"	"	
			100	"	"	"	1 V	"		"	"	F75	"		"	"	
			101	"	"	"	2 V	"		"	"	F76	"	N = 73 to 84	"	"	
			102	"	"	"	3 V	"		"	"	F77	"		"	"	
			103	"	"	"	4 V	"		"	"	F78	"		"	"	
			104	"	"	"	5 V	"		"	"	F79	"		"	"	
			105	"	"	"	6 V	"		"	"	F80	"		"	"	
			106	"	"	"	7 V	"		"	"	F81	"		"	"	
			107	"	"	"	8 V	"		"	"	F82	"		"	"	
			108	"	"	"	9 V	"		"	"	F83	"		"	"	
			109	"	"	"	10 V	"		"	"	F84	"		"	"	
	V_{OS1}		110	f = 10K	"	"	.01 V	"		K2	"	F61	"	V_{OS1} = .01-F61*9.99/(F72-F61)	-0.6	0.6	
				f = 10K	"	"	10 V	"		K2	"	F62	"				mV
	V_{OS2}		111	f = 1M	"	"	.01 V	"		K1,K5	"	F73	"	V_{OS2} = .01-F73*9.99/(F84-F73)	-0.6	0.6	
				f = 1M	"	"	10 V	"		K1,K5	"	F84	"				
	A_E		112	f = 10K	"	"	10 V	"		K2	"	F72	"	A_{E1} = .01((F72-F61)x10/9.99-10K)	-10	10	
			113	f = 1M	"	"	"	"		K1,K5	"	F84	"	A_{E2} = .0001((F84-F73)x10/9.99-1M)	-10	10	% of FSR
	dV_{OS}/dT		114	f = 10K										dV_{OS}/dT = V_{OS1}(-55°C)-V_{OS1}(+25°C)/80	-5	5	uV/°C
			115	f = 1M										dV_{OS}/dT = V_{OS2}(-55°C)-V_{OS2}(+25°C)/80	-5	5	
	d_{AE}/dT		116	f = 10K										d_{AE1}/dT = A_{E1}(-55°C)-A_{E1}(+25°C)/80	-50	50	ppm/°C
			117	f = 1M										d_{AE2}/dT = A_{E2}(-55°C)-A_{E2}(+25°C/80	"	"	
	$d_{F(FS)}/dT$		118	f = 10K							6	F72	Hz	$d_F(FS1)/dT$ = FS1)(-55°C)-FS1(+25°C)/80	"	"	ppm of FSR/°C
			119	f = 1M							6	F84	Hz	$d_F(FS2)/dT$ = FS2(-55°C)-FS2(+25°C)/80	"	"	

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TABLE III. Group A inspection for device type 03.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Notes	Applied voltages and currents ref. pin numbers (see figure 4) 1	2	3	4	5	Energized relays	Measured pin No.	Value	Unit	Equation	Limits Min	Max	Unit
1 T_A = +25°C	V_{OL1}		1		15 V	-15 V	0 V	8 mA			6	E1	V	V_{OL1} = E1	0	0.4	V
	V_{OL2}		2		15 V	-15 V	0 V	16 mA			6	E2	V	V_{OL2} = E2	0	0.8	
	I_{OH}		3						17 V	K4	5	I1	mA	I_{OH} = I1		100	µA
	I_{CC}		4		15 V	-15 V	0 V			None	1	I2	"	I_{CC} = I2	0	8.0	mA
	I_{EE}		5		"	"	0 V			None	2	I3	"	I_{EE} = I3	-8.0	0	
	L_E		6	f = 10K	"	"	.01 V	5 V			6	F1	Hz	L_E = .01(F(N)-(F12*(V_{IN}-V_{OS1}))/(10-V_{OS1}))	.005	.005	
			7	"	"	"	.1 V	"			"	F2	"		"	"	
			8	"	"	"	1 V	"			"	F3	"		"	"	
			9	"	"	"	2 V	"			"	F4	"	N = 1 to 12	"	"	
			10	"	"	"	3 V	"			"	F5	"		"	"	
			11	"	"	"	4 V	"			"	F6	"		"	"	
			12	"	"	"	5 V	"			"	F7	"		"	"	
			13	"	"	"	6 V	"			"	F8	"		"	"	
			14	"	"	"	7 V	"			"	F9	"		"	"	
			15	"	"	"	8 V	"			"	F10	"		"	"	
			16	"	"	"	9 V	"			"	F11	"		"	"	
			17	"	"	"	10 V	"			"	F12	"		"	"	% of FSR
			18	f = 1M	"	"	.01 V	"		K1,K2,K6	"	F13	"	L_E = .0005(F(N)-(F24*(V_{IN}-V_{OS2}))/(10-V_{OS2}))	0.1	0.1	
			19	"	"	"	.1 V	"		"	"	F14	"		"	"	
			20	"	"	"	1 V	"		"	"	F15	"		"	"	
			21	"	"	"	2 V	"		"	"	F16	"	N = 13 to 24	"	"	
			22	"	"	"	3 V	"		"	"	F17	"		"	"	
			23	"	"	"	4 V	"		"	"	F18	"		"	"	
			24	"	"	"	5 V	"		"	"	F19	"		"	"	
			25	"	"	"	6 V	"		"	"	F20	"		"	"	
			26	"	"	"	7 V	"		"	"	F21	"		"	"	
			27	"	"	"	8 V	"		"	"	F22	"		"	"	
			28	"	"	"	9 V	"		"	"	F23	"		"	"	
			29	"	"	"	10 V	"		"	"	F24	"		"	"	
	V_{OS1}		30	f = 10K	"	"	.01 V	"			"	F1	"	V_{OS1} = .01+I1*9.99/(F12-F1)	-4	4	mV
				f = 10K	"	"	10 V	"			"	F12	"				
	V_{OS2}		31	f = 1M	"	"	.01 V	"		K1,K2,K6	"	F13	"	V_{OS2} = .01+I3*9.99/(F24-F13)	-4	4	mV
				f = 1M	"	"	10 V	"		K1,K2,K6	"	F24	"				
	$+I_{IB}$	4001	32	f = 10K	"	"	.2 V	"		K12	"	F25	"	V_{OS3} = .2-F25*9.8/(F26-F25)	-100	100	nA
				"	"	"	10 V	"		K12	"	F26	"	$+I_{IB}$=(V_{OS3}-V_{OS1})/10K			
	$-I_{IB}$	4001	33	"	"	"	.1 V	"		K10	"	F27	"	V_{OS4} = .1+F27*9.9/(F28-F27)	-20	20	nA
				"	"	"	10 V	"		K10	"	F28	"	$+I_{IB}$=(V_{OS4}-V_{OS1})/10K			
	A_E		34	"	"	"	"	"		K1,K2,K6	"	F12	"	A_{E1} = .01((F12-F1)x10/9.99-10K)	-15	15	% of FSR
			35	f = 1M	"	"	"	"		K1,K2,K6	"	F24	"	A_{E2} = .0001((F24-F13)x10/9.99-1M)	-15	15	% of FSR
	P_{SS1}		36	f = 10K	11 V	"	"	"		K1,K2,K6	"	F29	"	P_{SS1} = 2.5(F29-F30)/F11	-.02	.02	% of FSR /% supply
			37	f = 10K	17 V	"	"	"			"	F30	"				
			38	f = 1M	11 V	"	"	"		K1,K2,K6	"	F31	"	P_{SS1} = 2.5(F31-F32)/F22	-.02	.02	
			39	f = 1M	17 V	"	"	"		K1,K2,K6	"	F32	"				

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Notes	Applied voltages and currents ref. pin numbers (see figure 4) 1	2	3	4	5	Energized relays	Measured pin No.	Measured pin Value	Measured pin Unit	Equation	Limits Min	Limits Max	Unit
1 T_A = +25°C	P_{SS2}		40	f = 10K	15 V	-11 V	10 V	5 V			6	F33	Hz	$P_{SS2} = 2.5(F33-F34)/F11$	-.02	.02	% of FSR
			41	f = 10K	"	-17 V	"	"			"	F34	"				
			42	f = 1M	"	-11 V	"	"		K1,K2,K6	"	F35	"	$P_{SS2} = 2.5(F35-F36)/F22$	-.02	.02	/% supply
			43	f = 1M	"	-17 V	"	"		K1,K2,K6	"	F36	"				
	F_{SE}		44	f = 10K	"	-15 V	"	"		K2,	"	F12	"	$F_{SE1} = F12$	-10	10	% of FSR
			45	f = 1M	"	"	"	"		K1,K6	"	F24	"	$F_{SE2} = F24$	-15	15	/% supply
	BpV_{OS1} BpV_{OS2}		46	f = 10K	"	"	"	"		K9,K13	"	F37	"	$BpV_{OS1} = F37$	-500	500	Hz
			47	f = 1M	"	"	"	"		K9,K13,K1, K2,K6	"	F38	"	$BpV_{OS2} = F38$	-50	50	kHz
2 T_A = +125°C	V_{OL1}		48		"	"	0 V	5 mA			6	E3	V	$V_{OL1} = E3$	0	0.4	V
	I_{CC}	3005	49		"	"	"			None	1	I4	mA	$I_{CC} = I4$	0	8.0	mA
	I_{EE}	3005	50		"	"	"	"		None	2	I5	mA	$I_{EE} = I5$	-8.0	0	mA
	L_E		51	f = 10K	"	"	.01 V	5 V			6	F39	Hz	$L_E = .01(F(N)-(F50*(V_{IN}-V_{OS1}))/(10-V_{OS1})$	-.005	.005	
			52	"	"	"	.1 V	"			"	F40	"		"	"	
			53	"	"	"	1 V	"			"	F41	"		"	"	
			54	"	"	"	2 V	"			"	F42	"	N = 39 to 50	"	"	
			55	"	"	"	3 V	"			"	F43	"		"	"	
			56	"	"	"	4 V	"			"	F44	"		"	"	
			57	"	"	"	5 V	"			"	F45	"		"	"	
			58	"	"	"	6 V	"			"	F46	"		"	"	
			59	"	"	"	7 V	"			"	F47	"		"	"	
			60	"	"	"	8 V	"			"	F48	"		"	"	
			61	"	"	"	9 V	"			"	F49	"		"	"	
			62	"	"	"	10 V	"			"	F50	"		"	"	% of FSR
			63	f = 1M	"	"	.01 V	"		K1,K2,K6	"	F51	"	$L_E = .0005(F(N)-(F62*(V_{IN}-V_{OS2})/(10-V_{OS2})$	-0.1	0.1	
			64	"	"	"	.1 V	"			"	F52	"		"	"	
			65	"	"	"	1 V	"			"	F53	"		"	"	
			66	"	"	"	2 V	"			"	F54	"	N = 51 to 62	"	"	
			67	"	"	"	3 V	"			"	F55	"		"	"	
			68	"	"	"	4 V	"			"	F56	"		"	"	
			69	"	"	"	5 V	"			"	F57	"		"	"	
			70	"	"	"	6 V	"			"	F58	"		"	"	
			71	"	"	"	7 V	"			"	F59	"		"	"	
			72	"	"	"	8 V	"			"	F60	"		"	"	
			73	"	"	"	9 V	"			"	F61	"		"	"	
			74	"	"	"	10 V	"			"	F62	"		"	"	
	V_{OS1}		75	f = 10K	"	"	.01 V	"			"	F39	"	$V_{OS1} = .01-F39*9.99/(F50-F39)$	-4.0	4.0	mV
				f = 10K	"	"	10 V	"			"	F50	"				
	V_{OS2}		76	f = 1M	"	"	.01 V	"		K1,K2,Ko	"	F51	"	$V_{OS2} = .01-F51*9.99/(F62-F51)$	-4.0	4.0	
				f = 1M	"	"	10 V	"		K1,K2,Ko	"	F62	"				
	A_E		77	f = 10K	"	"	"	"			"	F50	"	$A_{E1} = .01((F50-F39)x10/9.99-10K)$	-10	10	% of FSR
			78	f = 1M	"	"	"	"			"	F62	"	$A_{E2} = .0001((F62-F51)x10/9.99-1M)$	-15	15	
	dV_{OS}/dT		79	f = 10K										$dV_{OS}/dT = V_{OS1}(+125°C)-V_{OS1}(+25°C))/100$	-30	30	µV/°C
			80	f = 1M							"	"	"	$dV_{OS}/dT = V_{OS2}(+125°C)-V_{OS2}(+25°C))/100$	-30	30	

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Notes	Applied voltages and currents ref. pin numbers (see figure 4) 1	2	3	4	5	Energized relays	Measured pin No.	Value	Unit	Equation	Limits Min	Max	Unit
2 T_A = +125°C	d_{AE}/dT		81	f = 10K										$d_{AE1}/dT = A_{E1}(+125°C)-A_{E1}(+25°C)/100$	-150	150	ppm/°C
			82	f = 1M										$d_{AE2}/dT = A_{E2}(+125°C)-A_{E2}(+25°C/100$	-350	350	
	$d_F(FS)/dT$		83	f = 10K							6	F50	Hz	$d_F(FS1)/dT = FS1(+125°C)-FS1(+25°C)/100$	-75	75	ppm of FSR°C
			84	f = 1M							"	F62	"	$d_F(FS2)dT = FS2(+125°C)-FS2(+25°C)/100$	-150	150	
	$d_{BP}VOS/dT$		85	f = 10K						K9,K13	"	F63	"	$d_{BP}VOS/dT = (B_p(VOS1(+125°C)-B_pVOS1(+25°C))(10-4)$	-.02	.02	% FSC/°C
			86	f = 1M						K1,K2,K6, K9,K13	"	F64	"	$d_{BP}VOS/dT = (B_p(VOS2(+125°C)-B_pVOS2(+25°C))(10-4)$	-.02	.02	
3 T_A = -55°C	(All test parameters, conditions, equations, and test limits are identical with those specified in table III, subgroup 2, T_A = +125°C. Test numbers 48 through 86.)																

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3.6.1 Serialization. All class S devices shall be serialized in accordance with MIL-M-38510.

3.6.2 Correctness of indexing and markings. All devices shall be subjected to the final electrical tests subgroup 1 specified in table II after part marking to verify that they are correctly indexed and identified by part number. Optionally, an approved electrical test may be devised especialy for this requirement.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 56 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in (method 1015 of MIL-STD-883).

(1) Class S devices: Dynamic tests (test condition D) using the circuit shown on figure 3. Test duration shall be as defined in MIL-STD-883.

(2) Class B devices: Test condition B or D, using the circuit shown on figure 3.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) for class S and B devices shall be as specified in MIL-M-38510, based on failures from group A, subgroup 1 test after cooldown as final electrical tests in accordance with method 5004 of MIL-STD-883 and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Test shall be as specified in table II herein.

b. Subgroups 4, 5, 6, 7, 8, 9, 10, and 11 shall be omitted.

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4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. Electrical parameters shall be as specified in table II herein. For class S devices, delta limits shall apply only to subgroup 1 of group B inspection.

b. Steady-state life test for class S devices shall be in accordance with table IIa of method 5005 of MIL-STD-883, using the circuit shown on figure 3.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection for class B devices.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition B, using the circuit shown on figure 3, or test condition D, using the circuit shown on figure 3.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. This group shall be measured only for initial qualification and thereafter for group D tests at +25°C.

TABLE IV. Group C end-point limits. T_A = 25°C.

Table III test no.	Device type	Symbol	Delta 1/		End point		Units
			Min	Max	Min	Max	
30	01	V_{OS}	-0.6	0.6	-4	4	mV
"	02	"	-0.25	0.25	-0.5	0.5	"
"	03	"	-0.5	0.5	-4	4	"
32	01	$+I_{IB}$	-15	15	-150	150	nA
"	02	"	-15	15	-30	30	"
"	03	"	-10	10	-150	150	"
33	01	$-I_{IB}$	-9	9	-90	90	nA
"	02	"	-4	4	-8	8	"
"	03	"	-5	5	-90	90	"
34	01	A_E	-1.0	1.0	-10	10	%
"	02	"	-1.0	1.0	-10	10	"
"	03	"	-1.0	1.0	-15	15	"

1/ Delta limits apply to the measured value (see delta definition in MIL-M-38510.

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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables. Electrical tests circuits as precribed herein or in the referenced test methods of MIL-STD-883 shall be acceptable. Other test circuits shall require the approval of the qualifying activity.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

4.5.2 Life test and burn-in cooldown procedure. When devices are measured at 25°C following application of steady-state life or burn-in test condition, they shall be cooled to within 10°C of their power stable condition at room temperature prior to the removal of the bias.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for lead lengths, or lead forming, if applicable. These requirements shall not affect the part number.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510 and MIL-STD-1331.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

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6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	VFC32
02	VFC320
03	AD650

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/138	MICROCIRCUITS, LINEAR, VOLTAGE-TO-FREQUENCY CONVERTS, MONOLITHIC SILICON

NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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